5.6 Reaction Kinetics (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	5. Physical Chemistry (A Level Only)
Topic	5.6 Reaction Kinetics (A Level Only)
Difficulty	Easy

Time allowed: 30

Score: /24

Percentage: /100

Question 1a

The rate equation for the reaction between reactants **X**, **Y** and **Z** is shown below.

$$\text{rate} = k\,[\mathbf{X}]^2\,[\mathbf{Y}]$$

State the orders with respect to **X**, **Y** and **Z**.

[3 marks]

Question 1b

State the overall order of this reaction.

[1 mark]

Question 1c

The rate of reaction is 3.72×10^{-5} mol dm^{-3} s^{-1} when the

- concentration of **X** is 0.01 mol dm^{-3}
- concentration of **Y** is 0.02 mol dm^{-3}
- concentration of **Z** is 0.04 mol dm^{-3}

Calculate the rate constant, *k*, for this reaction, including the units.

[3 marks]

Question 1d

The experiment was repeated but the initial concentration of **X** was doubled, all other concentrations remained the same.

State what the effect would be on the rate of reaction.

Explain your answer.

[2 marks]

Question 2a

Bromide ions and bromate(V) ions react in acidic conditions to form aqueous bromine in the following equation.

$$5Br^- (aq) + BrO_3^- (aq) + 6H^+ (aq) \rightarrow 3Br_2 (aq) + 3H_2O (l)$$

A series of experiments was carried out at a given temperature to find the rate equation for the reaction.

The results from experiments using different hydrogen ion concentrations are shown in Fig. 2.1.

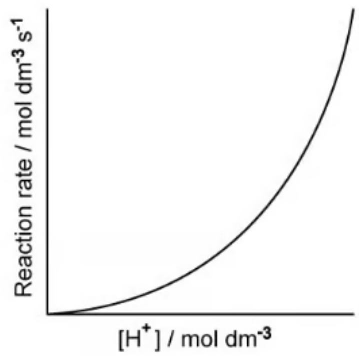


Fig. 2.1

Use the information in Fig. 2.1 to determine the order of reaction with respect to hydrogen ions.

[1 mark]

Question 2b

Experiments using different bromide concentrations showed that the order of reaction with respect to bromide ions was first order.

On the graph in Fig. 2.2, sketch a graph to show how the concentration of bromide ions would change during the course of a reaction.

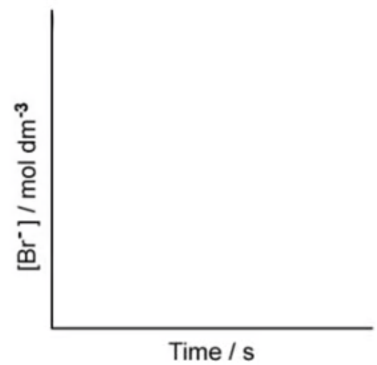


Fig. 2.2

[1 mark]

Question 2c

Overall, the reaction between bromide ions and bromate(V) ions in acidic conditions is fourth order.

i)

Deduce the order with respect to bromate(V) ions.

ii)

Using your answer to part **(a)** and the information in part **(b)**, write the rate equation for the reaction.

[2 marks]

Question 2d

Bromide ions and bromate(V) ions react in acidic conditions to form aqueous bromine in the following equation.

$$5Br^- (aq) + BrO_3^- (aq) + 6H^+ (aq) \rightarrow 3Br_2 (aq) + 3H_2O (l)$$

i)

Using your answer to part **(c)**, write an expression for the rate constant for the reaction between bromide ions and bromate(V) ions in acidic conditions.

ii)

Suggest suitable units for the rate constant.

[2 marks]

Question 3a

Hydrogen peroxide, H_2O_2, is a colourless liquid. It is widely used in cosmetic and medical products. It is an effective disinfectant and bleaching agent.

Hydrogen peroxide is unstable and will decompose slowly to form water and oxygen. The rate of decomposition can be increased using manganese(IV) oxide as a catalyst.

$$2H_2O_2\,(aq) \rightarrow 2H_2O\,(l) + O_2\,(g)$$

The graph in Fig. 3.1 shows how hydrogen peroxide decomposes in the presence of manganese(IV) oxide.

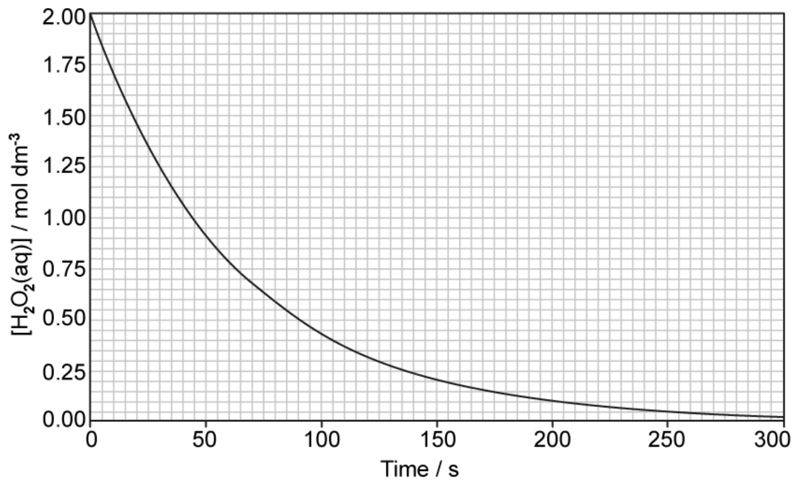


Fig. 3.1

State what is meant by the half-life of a reaction.

[1 mark]

Question 3b

Use the graph in Fig. 3.1 to show that this reaction is first order with respect to hydrogen peroxide.

Deduce the rate equation of the reaction.

[3 marks]

Question 3c

Using the graph in Fig. 3.2, determine

i)
the rate of reaction, in mol dm^{-3} s^{-1}, at 100 seconds.

ii)
the rate constant for this reaction. State the units.

Your answer must show full working on the graph in Fig 3.2.

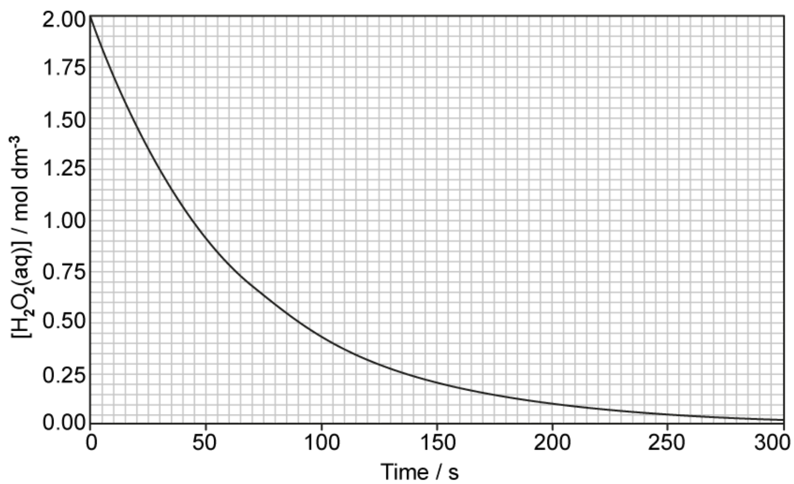


Fig. 3.2

[4 marks]

Question 3d

When the initial concentration of hydrogen peroxide was halved, state what the effect, if any, on the half-life of this reaction would be.

[1 mark]

Model Answers: Easy

1a

a) The orders with respect to each reactant are:

- **X** - 2 / 2nd / second order; [1 mark]
- **Y** - 1 / 1st / first order; [1 mark]
- **Z** - 0 / zero order; [1 mark]

[Total: 3 marks]

- The order of a reaction is found from the rate equation by looking at the power to which the concentration of that reactant is raised in the rate equation
- In the rate equation, [**X**] is raised to the power of two, therefore it is second order with respect to **X**
- [**Y**] is raised to the power of 1, therefore it is first order with respect to **Y**
  - $[\mathbf{Y}]^1 = [\mathbf{Y}]$, so 1st order powers are usually not shown in the rate equation
- **Z** does not appear in the rate equation, so it must be raised to the power of zero, therefore it is zero order with respect to **Z**
  - $[\mathbf{Z}]^0 = 1$, so zero order reactants are usually not shown in the rate equation
- **Tip**: it is easy to forget to include the order with respect to **Z** as it doesn't appear in the rate equation, this is where reading the question thoroughly is important

1b

b) The overall order of this reaction is:

- 3 / 3rd / third (order); [1 mark]

[Total: 1 mark]

- The overall order of reaction is the sum of orders with respect to each reactant
- The reaction is 2nd order with respect to **X**; 1st order with respect to **Y** and zero order with respect to **Z**
- The overall order is therefore 2 + 1 + 0 = 3

1c

c) The rate constant with units is:

- $k = \dfrac{3.72 \times 10^{-5}}{(0.01)^2 \times 0.02}$; [1 mark]
- $k = 18.6$; [1 mark]
- Units: $dm^6\ mol^{-2}\ s^{-1}$; [1 mark]

[Total: 3 marks]

- Calculating the rate constant is simply a matter of substituting the values given into the rate equation and rearranging it to give *k* as the subject
- **Careful**: It is a common mistake for students to rearrange incorrectly; if you are not confident with rearranging equations, it can be better to substitute the values into the rate equation and simplify before rearranging, as follows:
  - Substituting values into the rate equation:

$$3.72 \times 10^{-5} = k \times (0.01)^2 \times 0.02$$

  - Simplifying:

$$3.72 \times 10^{-5} = k \times 2.00 \times 10^{-6}$$

  - Rearranging:

$$k = \frac{3.72 \times 10^{-5}}{2.00 \times 10^{-6}} = 18.6$$

- Although you are given the concentration of **Z**, you do not need to use it in this calculation

- For all rate constant questions, you can be expected to deduce / suggest suitable units
  - Substitute in the units in the rate constant expression:

$$k = \frac{\text{rate}}{[X]^2\,[Y]} \rightarrow k = \frac{\text{mol dm}^{-3}\ \text{s}^{-1}}{(\text{mol dm}^{-3})^2\ (\text{mol dm}^{-3})}$$

  - Cancel out / simplify

$$k = \frac{\text{mol dm}^{-3}\ \text{s}^{-1}}{(\text{mol dm}^{-3})^2\ (\text{mol dm}^{-3})} \rightarrow k = \frac{\text{s}^{-1}}{(\text{mol dm}^{-3})^2}$$

- Write the final answer:

$$k = \frac{\text{s}^{-1}}{(\text{mol dm}^{-3})^2} \rightarrow k = \text{dm}^6\ \text{mol}^{-2}\ \text{s}^{-1}$$

- It is the convention to write positive indices before negative indices but you will not get penalised in an exam if you do not do this
- Common mistakes include:
  - Writing the expression upside down, this would give a calculated value of 0.0538 and would score 1 mark
    - If you then gave the units of mol^2 s^1 dm^{-6} you would gain an 'error carried forward' mark, i.e. you are not penalised a second time for the previous error
  - Not squaring 0.01, this would give a calculated value of 0.186 and would score 1 mark
    - This error would give the units to be dm^3 mol^{-1} s^{-1} which would also be an ECF mark (error carried forward)

1d

d) The effect on the rate of reaction is:

- (The rate of reaction would) quadruple / x4 / become 1.488 x 10^{-4} (mol dm^{-3} s^{-1}); [1 mark]
- (Because) the order is 2 with respect to **X**; [1 mark]

[Total: 2 marks]

- As the reaction is 2nd order with respect to **X**, as the concentration of **X** doubles, the rate will increase by a factor of 2^2 = 4
- The initial rate of reaction was 3.72 x 10^{-5} so if it increases by a factor of 4, it becomes 3.72 x 10^{-5} x 4 = 1.488 x 10^{-4} mol dm^{-3} s^{-1}
- A value of 1.5 x 10^{-4} to the calculator value would be accepted

2a

a) The order of reaction with respect to hydrogen ions is:

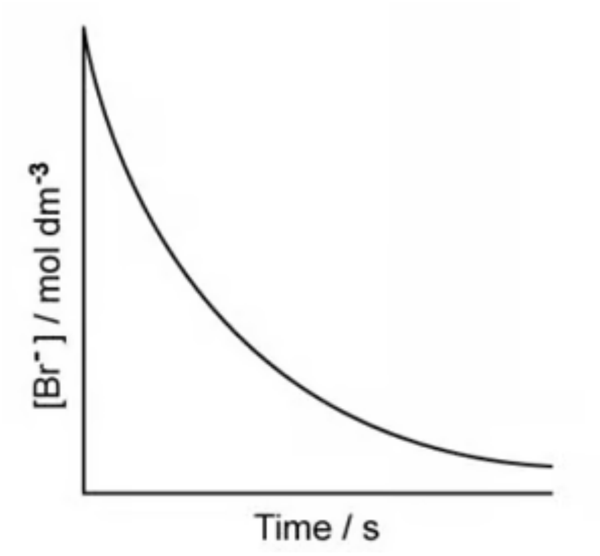
- 2 / 2nd / second order; [1 mark]

[Total: 1 mark]

- You should be able to recognise the orders for each reactant from rate-concentration graphs
  - Reactants that are zero order produce a horizontal / constant line on a rate-concentration graph
    - This shows that the rate does not change with concentration
  - Reactants that are first order produce a straight line from (0,0) on a rate-concentration graph
    - This shows that shows that rate is directly proportional to concentration
  - Reactants that are second order produce a curve on a rate-concentration graph
    - This shows that rate increases exponentially with concentration,
- As the rate-concentration graph shows a curve, the order of reaction with respect to hydrogen ions must be second order / 2

2b

b) The concentration-time graph for bromide ions is:



- Shallow curve, decreasing as time increases; [1 mark]

[Total: 1 mark]

- Reactants that are zero order produce a **straight line**, that shows concentration decreasing as time increases, on a concentration-time graph
- Reactants that are first order produce a **shallow curve**, that shows concentration decreasing as time increases with a constant half-life on a concentration-time graph
- Reactants that are second order produce a **steep curve**, that shows concentration decreasing as time increases, on a concentration-time graph
- As the order of reaction with respect to bromide ions is first order, the concentration-time graph will show a **shallow curve**
- As this is a sketch, you are not expected to use half-lives to accurately determine the shape of the curve but it can be useful to roughly plot 3 points or so to help you get the approximate gradient of the curve
- An examiner is looking to see that the gradient isn't too steep and that it doesn't have a long 'tail' as it levels off - these are both features of a second order curve

2c

c)

i) The order with respect to bromate(V) ions is:

- 1 / 1st / first order; [1 mark]

ii) The rate equation for the reaction is:

- Rate = $k[Br^-][BrO_3^-]]H^+]^2$; [1 mark]

[Total: 2 marks]

- From part (a), you have deduced that the order of reaction for hydrogen ions is 2
- In part (b), you are given that the order of reaction for bromide ions is 1
- Therefore, the order of reaction with respect to bromate(V) ions **must** be 1
  - Bromide + bromate(V) + hydrogen = 4
  - 1 + bromate(V) + 2 = 4
  - bromate(V) = 4 - 1 - 2 = 1
- Rate equations are of the general format
  - Rate = $k[A]^m[B]^n$
  - Where [A] and [B] are the concentrations of reactants and m and n are the

- Reactants that are zero order produce a **straight line**, that shows concentration decreasing as time increases, on a concentration-time graph
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- As the order of reaction with respect to bromide ions is first order, the concentration-time graph will show a **shallow curve**
- As this is a sketch, you are not expected to use half-lives to accurately determine the shape of the curve but it can be useful to roughly plot 3 points or so to help you get the approximate gradient of the curve
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2c

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- 1 / 1st / first order; [1 mark]

ii) The rate equation for the reaction is:

- Rate = $k\,[Br^-]\,[BrO_3^-]\,]H^+]^2$; [1 mark]

[Total: 2 marks]

- From part (a), you have deduced that the order of reaction for hydrogen ions is 2
- In part (b), you are given that the order of reaction for bromide ions is 1
- Therefore, the order of reaction with respect to bromate(V) ions **must** be 1
  - Bromide + bromate(V) + hydrogen = 4
  - 1 + bromate(V) + 2 = 4
  - bromate(V) = 4 - 1 - 2 = 1
- Rate equations are of the general format
  - Rate = k $[A]^m\,[B]^n$
  - Where [A] and [B] are the concentrations of reactants and m and n are the respective orders of the reaction
  - Having deduced that the orders of reaction are: bromide ions = 1,

bromate(V) ions = 1 and hydrogen ions = 2, you can use the general format to write your final answer

2d

d)

i) The expression for the rate constant for the reaction between bromide ions and bromate(V) ions in acidic conditions is:

- $k = \frac{\text{rate}}{[Br^-]\,[BrO_3^-]\,[H^+]^2}$; [1 mark]

ii) Suitable units for the rate constant in part (a) are:

- $dm^9\ mol^{-3}\ s^{-1}$; [1 mark]

[Total: 2 marks]

- You are expected to be able to rearrange the rate constant expression that you deduce
- For all rate constant questions, you can be expected to deduce / suggest suitable units
- Substitute in the units in the rate constant expression:

$$k = \frac{\text{rate}}{[Br^-]\,[BrO_3^-]\,[H^+]^2} \rightarrow k = \frac{\text{mol dm}^{-3}\ \text{s}^{-1}}{(\text{mol dm}^{-3})\ (\text{mol dm}^{-3})\ (\text{mol dm}^{-3})^2}$$

- Cancel out / simplify

$$k = \frac{\text{mol dm}^{-3}\ \text{s}^{-1}}{(\text{mol dm}^{-3})\ (\text{mol dm}^{-3})\ (\text{mol dm}^{-3})^3} \rightarrow k = \frac{\text{s}^{-1}}{(\text{mol dm}^{-3})^3}$$

- Write the final answer:

$$k = \frac{\text{s}^{-1}}{(\text{mol dm}^{-3})^3} \rightarrow k = \text{dm}^9\ \text{mol}^{-3}\ \text{s}^{-1}$$

- It is the convention to write positive indices before negative indices but you will not get penalised in an exam if you do not do this

3a

a) The half-life of a reaction is:

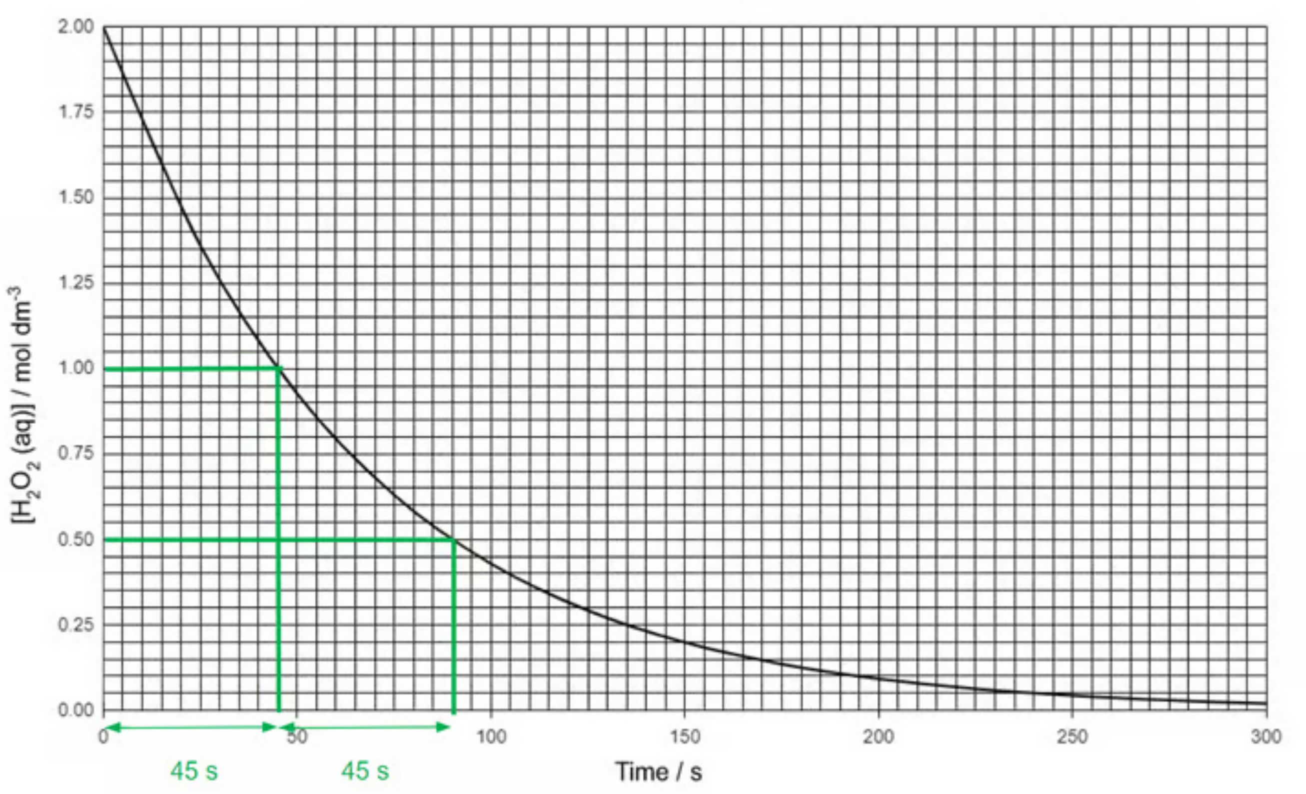
- The time taken for half of a reactant to be used up / react; [1 mark]

[Total: 1 mark]

- This is a definition that you are expected to know,

3b

b) To show that the reaction is first order with respect to hydrogen peroxide:



- Evidence from graph, either drawn or stated with 2 half-lives; [1 mark]
- Half-life: 45 ± 2 s (43–48); [1 mark]
- Rate equation: Rate = $k[H_2O_2]$; [1 mark]

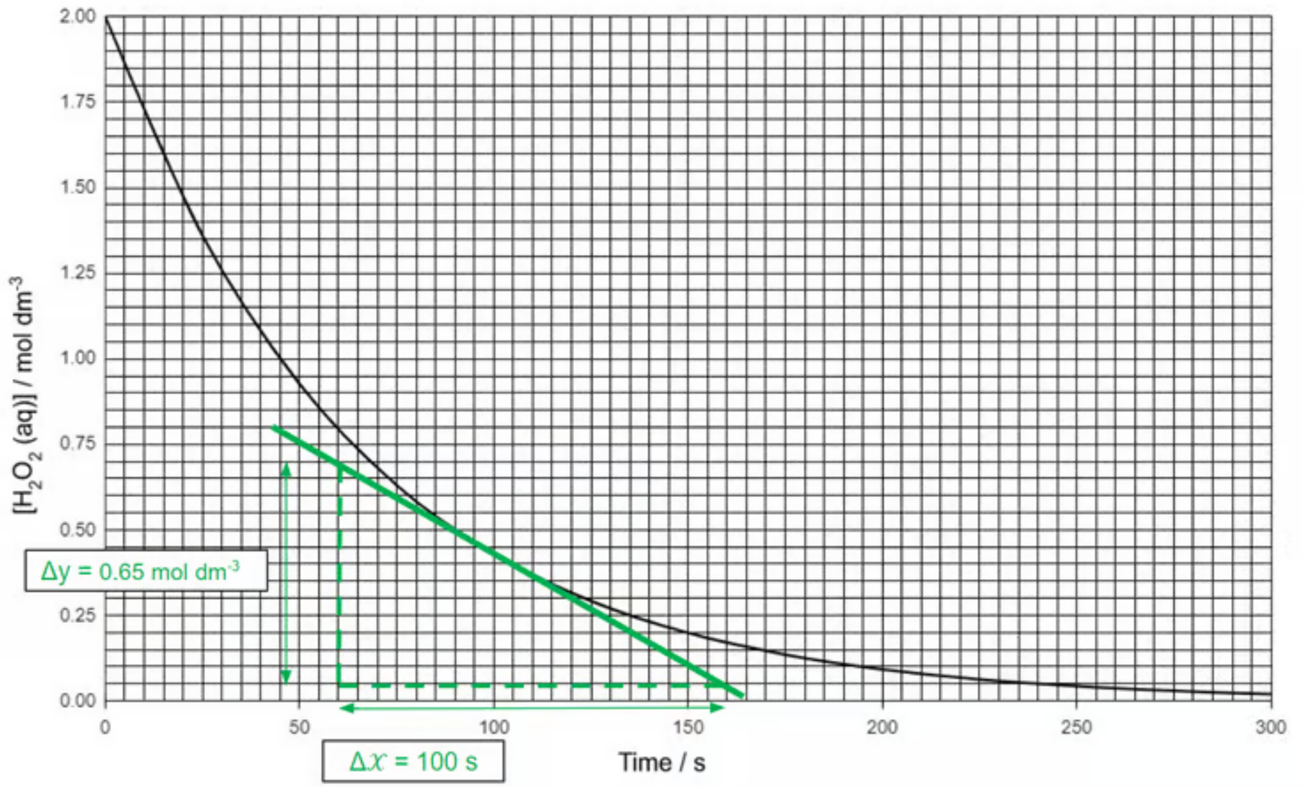
[Total: 3 marks]

- The question asks you to 'Use the graph' which suggests that you are expected to annotate the graph
- To show the reaction is first order, you must show that the half-life is constant, so draw at least 2 successive half-lives on the graph
- It is a common mistake for students to just find the half-life of the initial concentration falling to half its value; this does not show that it is a first order reaction as it doesn't show that the half-life is constant

3c

c)

i) The rate of reaction can be determined as follows:



- Evidence of tangent drawn on graph to line at $t = 100$ s; [1 mark]
- Gradient determined in range $6.5 \pm 0.5 \times 10^{-3}$ (mol dm^{-3} s^{-1}); [1 mark]

ii) The rate constant can be determined as follows:

- Rate constant clearly linked to rate, $k = \dfrac{\text{rate}}{[H_2O_2]}$

  OR

Rate constant clearly linked to half-life, $k = \frac{\ln 2}{t_{1/2}}$; [1 mark]

- Units: s^{-1}; [1 mark]

[Total: 4 marks]

- **Exam tip**: Draw tangents carefully and make sure to place the ruler right up against the curve
  - The tangent is correctly positioned when the angles between the curve and the tangent on either side of the point that touches the curve are the same.
- Example calculations:
  - The gradient is calculated by $\frac{\Delta y}{\Delta x} = \frac{0.65}{100\ s} = 6.5 \times 10^{-3}\ mol\ dm^{-3}\ s^{-1}$
- Example calculations for determining *k* using the two different methods in the mark scheme:
  - Using the rate at t = 100 s as calculated from the graph, where concentration = 0.425 mol dm^{-3}
    - $k = \frac{rate}{[H_2O_2]}$ $\quad k = \frac{6.5 \times 10^{-3}\ mol\ dm^{-3}\ s^{-1}}{0.425\ mol\ dm^{-3}} = 0.0153\ s^{-1}$
  - Using the half-life
    - $k = \frac{\ln 2}{t_{1/2}}$ $\quad k = \frac{\ln 2}{45\ s} = 0.0154\ s^{-1}$
- You would be awarded the marks for a calculated gradient with a value ± 0.5 x 10^{-3}, i.e. values in range 6.0 x 10^{-3} to 7.0 x 10^{-3} mol dm^{-3} s^{-1}
- If you did not calculate the gradient correctly due to an incorrectly drawn tangent, you can still gain 2 marks in part (ii) using your calculated value

3d

d) The effect on the half-life if the concentration of hydrogen peroxide was halved is:

- No effect; [1 mark]

[Total: 1 mark]

- The half-life of a reaction is independent of the initial concentration and it remains constant over time for a first order reaction
- Changing the initial concentration therefore has no effect on the half-life of the